

COURSE TITLE	: ANALYTICAL & BIOMEDICAL INSTRUMENTS LAB
COURSE CODE	: 6218
COURSE CATEGORY	: A
PERIODS/WEEK	: 5
PERIODS/SEMESTER	: 75
CREDITS	: 3

On completion of this course the student will be able to

1. Measure the pH value of a solution using pH meter.
2. Determination of refractive index of glass prism using spectrometer.
3. Determination of refractive index of water and glass slab using travelling microscope.
4. Determination of the wavelength of light by Newton's ring method.
5. Study of Gas analyzers-thermal conductivity and electrical conductivity analyzer.
6. Determination of concentration of given sample using Spectrophotometer.
7. Study of Paramagnetic O2 analyzer
8. Study of Air pollution monitoring equipment
9. Measure conductivity of a solution using conductivity meter.
10. Measure blood pressure using Sphygmomanometer
11. Measure blood pressure using Digital BP apparatus
12. Demonstrate ECG machine
13. Demonstrate EEG machine.
14. Demonstrate EMG machine.
15. Demonstrate Nebulizers
16. Measure Blood sugar using Glucometer.
17. Determination of concentration of given sample using Photo colorimeter.
18. Determine concentration of given sample using Flame Photometer.
19. Setup pacemaker circuit and plot the output waveform.
20. Measure skin contact impedance using Electrodes
21. Setup Bio amplifier circuit using op-amp and find out CMRR.
22. Set up Precision Rectifier Circuit using OP amp
23. Demonstrate the characteristics of Electrodes and transducers.
24. To demonstrate the working of Bedside monitor.